HeatShift

Tomorrow’s temperature, with evidence you can check

Karachi, Lahore and Multan
A 24-hour airport temperature prototype

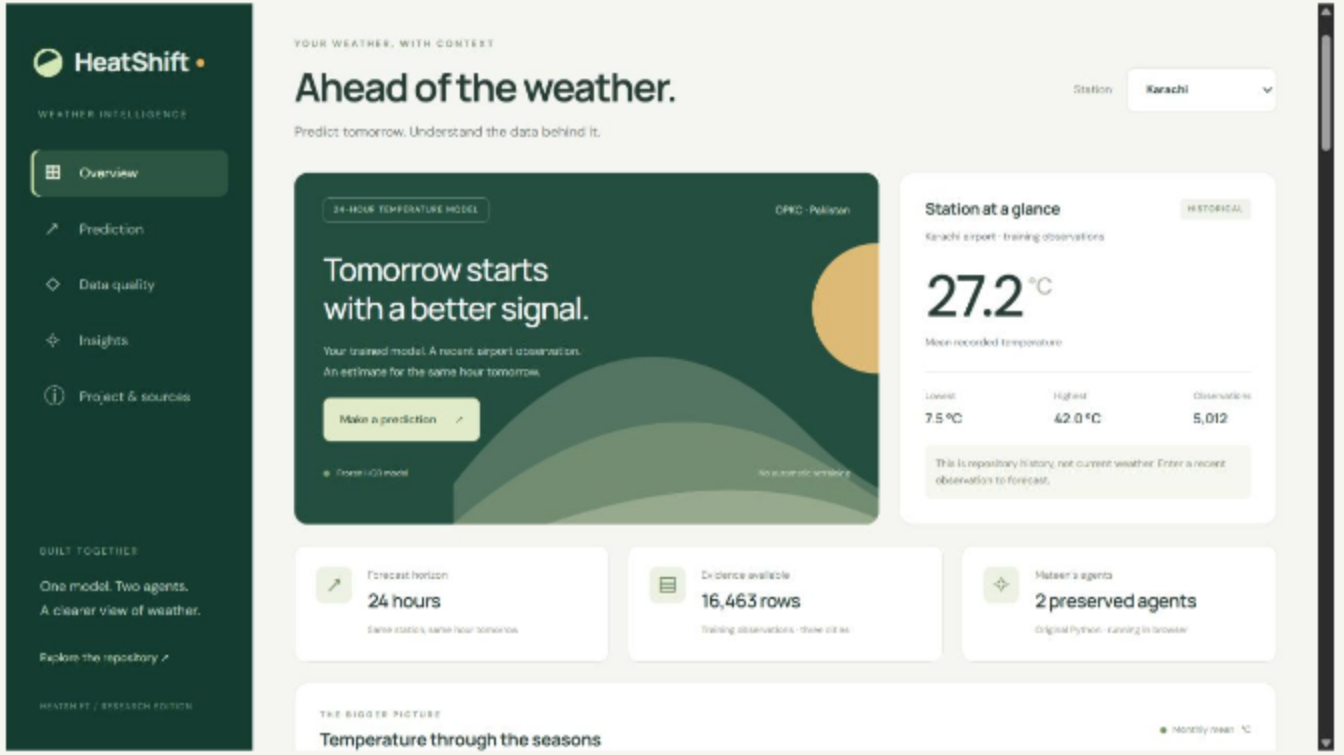
Waleed Irfan
Syed Mateen Ali
Aman Khan Lodhi

The problem and our solution

A forecast number leaves questions

How reliable are the inputs?
How uncertain is the estimate?
Where did the explanation come from?

Prediction, data checks and traceable answers in one view



Public responsive dashboard

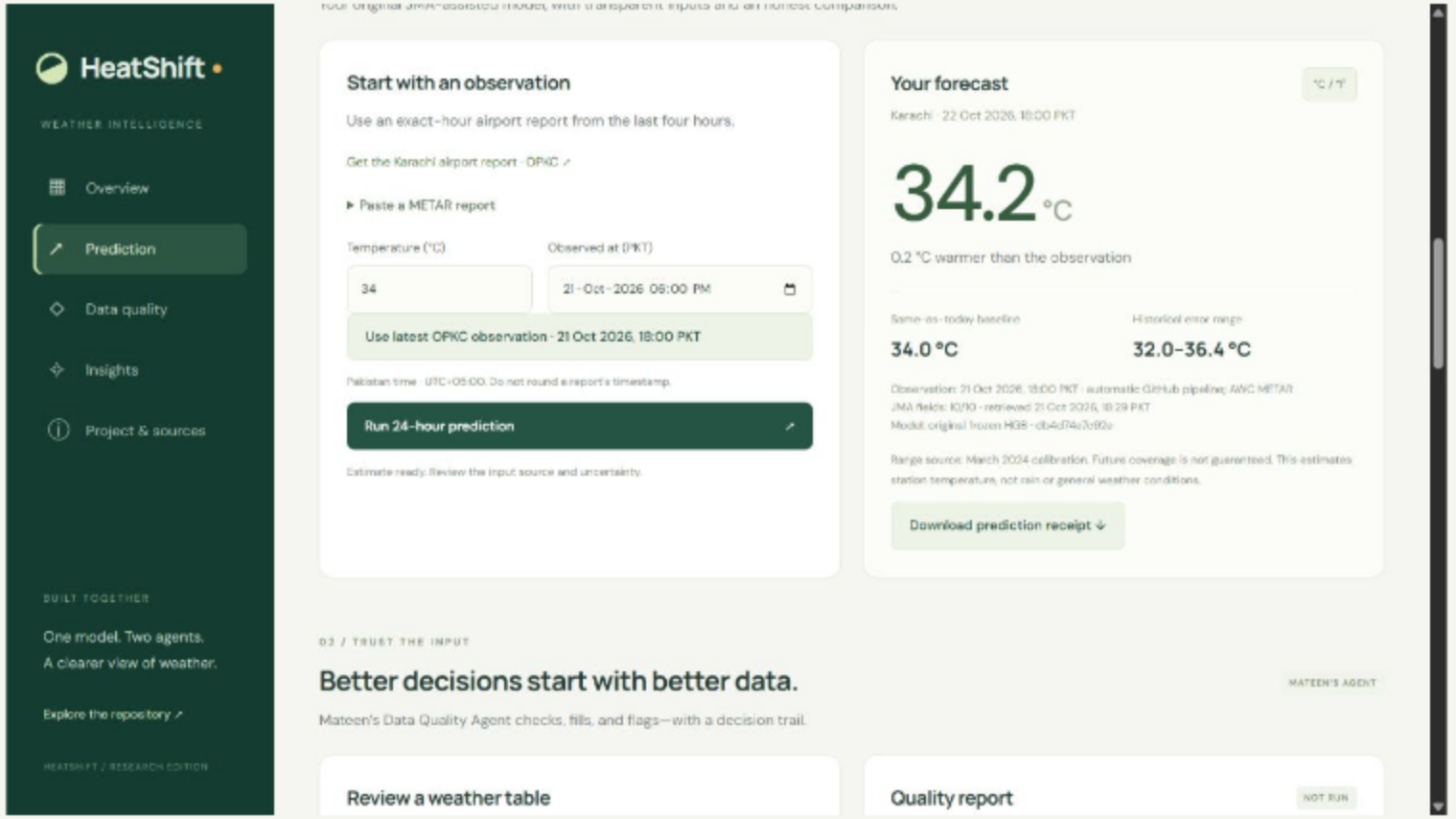
The data warehouse

Original records stay separate from cleaned and processed data

01	**Acquire**	NOAA station observations and archived weather inputs
02	**Preserve**	Raw snapshots and manifest.csv with source receipts
03	**Prepare**	Cleaned tables, exact 24-hour targets and time-based splits
04	**Rebuild**	One make data command and 14 warehouse checks

16,463 training rows 1,783 validation rows 755 calibration rows

The prediction model



Frozen
HistGradientBoosting

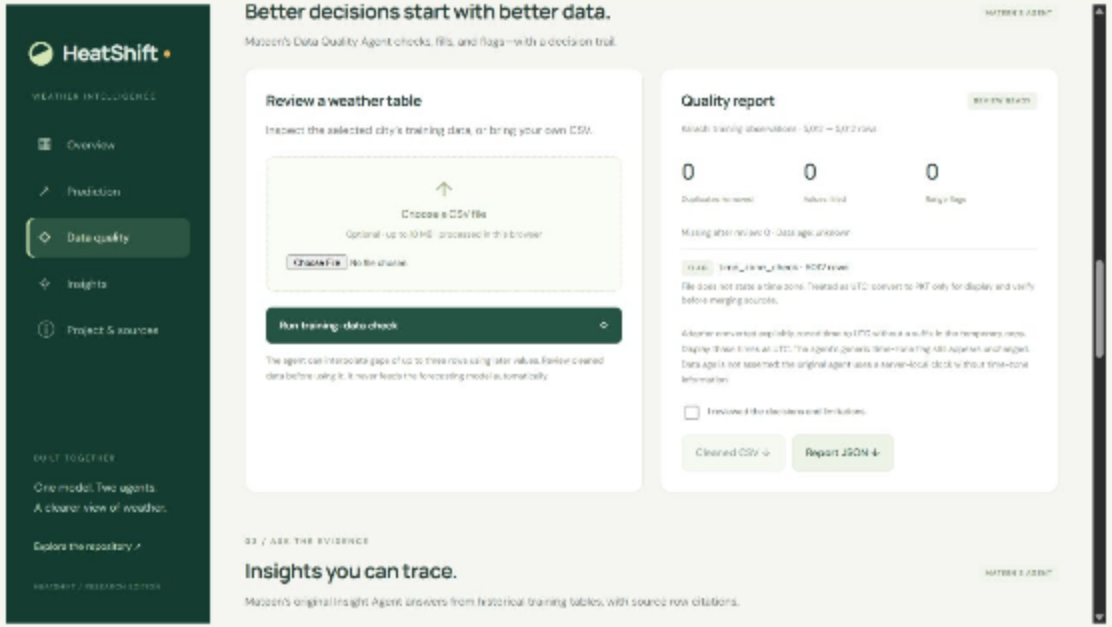
Recent station temperature
JMA weather forecast fields
Station and calendar features

Output
Station temperature
24 hours later

Example captured on 21 October 2026. The model does not retrain on each request.

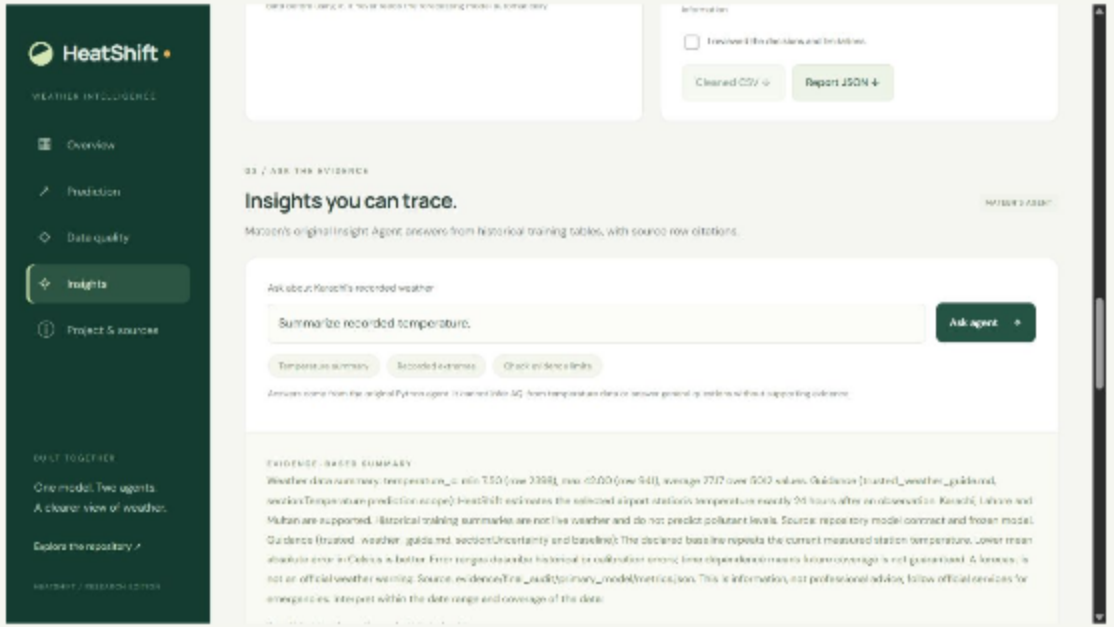
Mateen’s two agents

Data Quality Agent



Removes duplicates, fills small gaps and flags suspect values with a log

Insight Agent



Retrieves trusted notes and computes table summaries with row or section citations

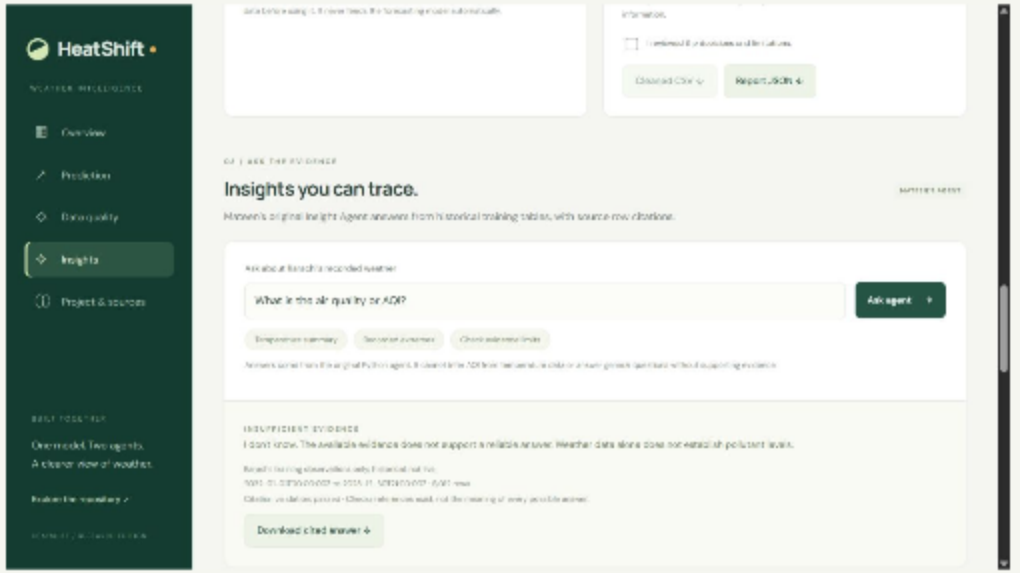
Live demonstration

1 Prediction and uncertainty

2 Quality decisions and downloadable log

3 A cited temperature summary

4 An unsupported AQI question



Open the public dashboard

Offline backup: the 2 min 24 sec demo video

Lower average validation error



10.8%

observed reduction in average absolute error

1,783 identical validation rows
Lower MAE is better

January performed worse than baseline. The improvement interval includes zero.
These reused validation results do not prove future performance.

Feasibility and business model

Who could benefit?

Everyday planners and students
Potential institutional payers: campuses, event teams and logistics firms

Basic public view, with paid reporting and support as a future hypothesis

What would it cost?

Current free static prototype
Hourly GitHub pipeline
Commercial weather-data rights
Maintenance and support

No verified paying customers or measured impact yet

Team contributions and questions

Waleed Irfan	Prediction model development and improvement
Syed Mateen Ali	Development, management and maintenance of both AI agents
Aman Khan Lodhi	Data cleaning, warehouse management and related preparation

Next: recover missing provenance and test future seasons independently

Backup: errors by group and uncertainty

Group	Rows	Model MAE	Baseline MAE
Karachi	419	1.031°C	1.368°C
Lahore	963	1.082°C	1.129°C
Multan	401	1.226°C	1.352°C
January	775	1.165°C	1.074°C
February	1,008	1.053°C	1.359°C

March calibration bands

Karachi ±2.22°C
Lahore ±3.08°C
Multan ±2.27°C

Nominal 90% ranges
Future coverage unknown

Baseline repeats the raw current NOAA temperature on the same scored rows.

Backup: effort, speed and memory

Measurement	Recorded result	Practical limit
Warm model inference	80.4 ms / 2,538 rows	Excludes loading and network
Node heap snapshot	11.0 MiB	Not total browser memory
Quality / insight runtime	0.238 s / 0.103 s	Local warm runs, 5,012 rows
Browser startup	About 25 MB download	First use can take a minute
Human effort	Exact hours not recorded	AI assistance disclosed

Earlier Ridge and station-only HGB attempts did not beat the baseline.
The final audit preserved the existing JMA-assisted model.

Backup: project links and evidence

Public HeatShift dashboard

GitHub source and submission documents

Public demo video

Evaluation metrics and row-level evidence

Primary data services

NOAA Integrated Surface Database
Open-Meteo previous-runs API
Aviation Weather Center METAR API